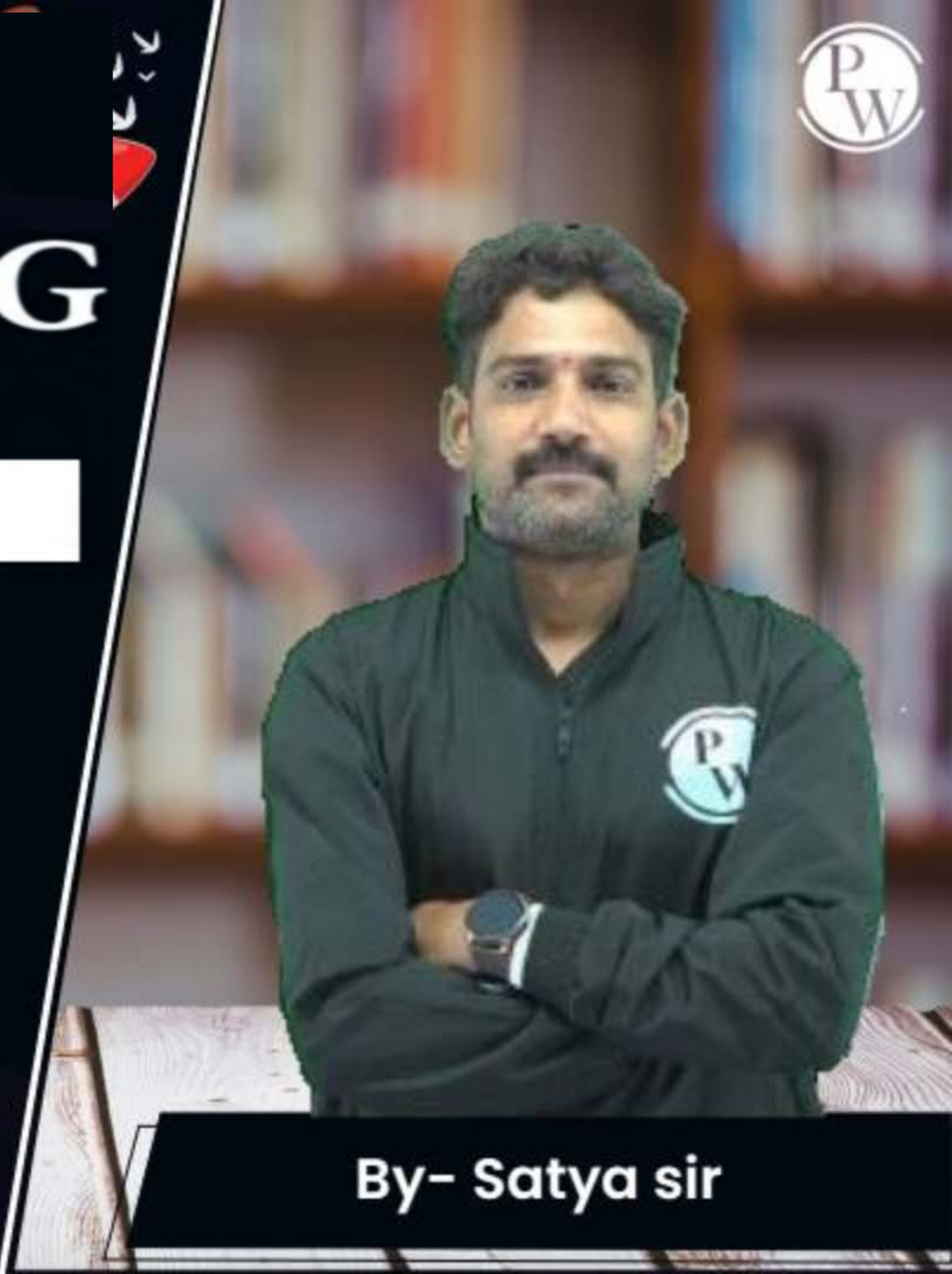
CS & IT ENGINEERING

Python For Data Science

Python Datatypes

DPP 01 (Discussion Notes)



By- Satya sir

[MCQ]

#Q. Output Printed by below Code is _____

```
Set1={2,4,6,8,10}
Set2={1,2,3,4,5}
Set3={1,3,5,7,9,x,x}
s1=Set1 & Set2
s2=Set2 | Set3
x=len(Set3)
for i in s1:
    for j in s2:
        x=x+1
print(x)
```

s1 = {2,4}

s2 = {1,2,3,4,5,7,9}

x = 5

i in s1

j in s2

	j=1	2	3	4	5	7	9
i=2	x=6	7	8	9	10	11	12
i=4	x=13	14	15	16	17	18	19

x=19

A 13

B 19

C 23

D 29

[NAT]

#Q. The number of times print statement is executed is 12

```
s1={5,7,9,7,5}
s2={3,5,7,5,3}
s3={1,2,3,4,5}
for i in range(len(s1.difference(s3))):
    for j in range(len(s3.union(s2))):
        print("Hi")
```

{7,9}

{1,2,3,4,5,7}

i = 0

j = 0,1,2,3,4,5 6 times

Print( )

i = 1

j = 0,1,2,3,4,5 6 times

Total Print() Stmt Executed is

6+6

=12 times

[MCQ]

#Q. The output of below code will be _____

```
d1={'a':1,'b':2,'c':3}
d2={'c':4,'d':5,'e':6}
d1.update(d2) print(d1)
```

d1 = {'a':1, 'b':2, 'c':4, 'd':5, 'e':6}

A {'a': 1, 'b': 2, 'c': 3, 'd': 5, 'e': 6}

B {'a': 1, 'b': 2, 'c' : 3,'c': 4, 'd': 5, 'e': 6}

C {'a': 1, 'b': 2, 'c': 4, 'd': 5, 'e': 6}

D {'a': 1, 'b': 2, 'd': 5, 'e': 6}

[NAT]

#Q. The final count value will be 133

```
myList=[11,22,33,44,55,66]
newList=[item for item in myList if item%2==0]
count=1
for i in range(len(newList)):
    count+=newList[i]
```

newList = [22, 44, 66]

i=0 count = 1 + 22 = 23

i=1 count = 23 + 44 = 67

i=2 count = 67 + 66 = 133

[MCQ]

#Q. What is printed by below code?

```
a={'apple','banana','mango'}
b={'mango','grapes','chiku'}
c={'chiku','banana','guava'}
x=len(a-b)
y=len(b^c)
z=len(c&a)
print(x+y+z)
```

a-b = {'apple', 'banana'}

b^c = {'mango', 'grapes', 'banana', 'guava'}

c&a = {'banana'}

x = 2

y = 4

z = 1

print(x+y+z) = 2+4+1 = 7

A 6

B 7 ✓

C 8

D 9

[NAT]

#Q. The total number of times print statement executed is 9

```
a=[10,20,[30,40,50],60,70]
b=(5,10,15,(20,25,30,(35,40,45,50),55,60),70)
c=a[2][1]
d=b[3][3][3]
i=1
for j in (c,1,c//4):
        for k in (d,1,c//5):
                print(i)
```

c = 40

d = 50

(40, 1, 40//4)

(50, 1, 50//5)

j	k	
	k=50	k=10	k=2	
j=40	3 times			3 times
j=10	3 times			3 times
j=2	3 times			3 times
				9 times

[MCQ]

#Q. The output will be ____

```
x="GATE"
y="2025"
z='EXAM'
string=x+y+z
print(len(string*2))
```

"GATE2025EXAM"

GATE2025EXAMGATE2025EXAM

len( )=24

A 4

B 8

C 16

D ✓ 24

[MCQ]

#Q. The ~~final value of ans~~ output will be _____

```
i={10,20,30,40,50}
j=[11,22,33,44,55]
k=[22,33,44,11,55]
l={30,20,40,10,50}
x=i == l
y=j != k
print(x,y)
```

- sets are unordered
- Lists are ordered.

x = i == l → True

y = j != k → True

A. False, False

B. True, False

C. False, True

D. ✓ True, True

THANK - YOU